

4.11. Secure Channel Protocol

This section specifies the formal protocol definition for the Secure Channel Protocol. Secure Channel Protocol defines the control plane for secure channel communication and security.

4.11.1. Secure Channel Protocol Messages

Secure Channel Protocol is composed of a collection of sub-protocols, including:

- Message Counter Synchronization Protocol (MCSP)
- Message Reliability Protocol (MRP)
- Passcode Based Session Establishment (PASE)
- Certificate Based Session Establishment (CASE)

The protocol opcodes for messages within the Secure Channel Protocol are grouped based on the underlying sub-protocol that uses the message type. [Table 18, “Secure Channel Protocol Opcodes”](#) lists the messages defined by Secure Channel Protocol.

Table 18. Secure Channel Protocol Opcodes

Protocol Opcode	Protocol Command Name	Description
Protocol ID = PROTOCOL_ID_SECURE_CHANNEL		
0x00	MsgCounterSyncReq	The Message Counter Synchronization Request message queries the current message counter from a peer to bootstrap replay protection.
0x01	MsgCounterSyncRsp	The Message Counter Synchronization Response message provides the current message counter from a peer to bootstrap replay protection.
0x10	MRP Standalone Acknowledgement	This message is dedicated for the purpose of sending a stand-alone acknowledgement when there is no other data message available to piggyback an acknowledgement on top of.
0x20	PBKDFParamRequest	The request for PBKDF parameters necessary to complete the PASE protocol.
0x21	PBKDFParamResponse	The PBKDF parameters sent in response to PBKDF-ParamRequest during the PASE protocol.
0x22	PASE Pake1	The first PAKE message of the PASE protocol.
0x23	PASE Pake2	The second PAKE message of the PASE protocol.
0x24	PASE Pake3	The third PAKE message of the PASE protocol.
0x30	CASE Sigma1	The first message of the CASE protocol.
0x31	CASE Sigma2	The second message of the CASE protocol.
0x32	CASE Sigma3	The third message of the CASE protocol.